

MATHEMATICS
Topics in Algebraic Geometry
Fall semester 2005

When: M,W,F 11:50-12:40PM

Where: LCB 215

Instructor: Prof. Christopher Hacon

Office: JWB 325

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Office hours: M, W, F 12.40–1.20 P.M. and by appointment.

Prerequisites: Some basic knowledge of Algebraic Geometry. Eg. you should be familiar with the words: projective variety, manifold, curve, surface, vector bundle, coherent sheaf, canonical line bundle, cohomology, divisors, Serre duality etc.

Topics: This course is intended to be an introduction to recent results and techniques used in the study of higher dimensional algebraic varieties. We will begin by discussing topics in the birational geometry of complex surfaces, then we will discuss the minimal model program developed by Mori, Kawamata, Kollár, Shokurov and others. Depending on the amount of time available and the interest of the students we will also discuss some of the more recent developments of the minimal model program and their applications to the birational geometry of algebraic varieties.

Books: I hope to have notes available for this class. I may follow some topics of:

Beauville, A.: *Complex algebraic surfaces*. Second edition. London Mathematical Society Student Texts, 34. Cambridge University Press, Cambridge, 1996.

Debarre, O.: *Higher-dimensional algebraic geometry*. Universitext. Springer-Verlag, New York, 2001.

Kollár, J.; Mori, S.: *Birational geometry of algebraic varieties*. With the collaboration of C. H. Clemens and A. Corti. Cambridge Tracts in Mathematics, 134. Cambridge University Press, Cambridge, 1998

Matsuki, K. *Introduction to the Mori program*. Universitext. Springer-Verlag, New York, 2002.